Could the class-number-one theorem have been proved in 1891?

A reconstruction from Weber's modular theory

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Abstract

We investigate a possible route to the imaginary quadratic class-number-one theorem using the modular theory in Weber's *Elliptische Functionen und algebraische Zahlen* (1891). The central deduction is Heegner's: comparing cubic equations for a singular value and its fourth power produces a Diophantine equation. We give that deduction, a complete elementary solution of the equation, and the bound $q < 196$, followed by a four-case exclusion above 163. We also give a specialization of Stark's repair of Weber's class-invariant argument, with explicit control of the roots of the replacement modular equation. The product construction used in this repair is already part of Weber's 1891 method. This supports 1891 as a historical upper-bound proposal under a stated rule permitting new applications of published methods. It neither dates Heegner's discovery to 1891 nor establishes the earliest possible year. Classical modular results imported from Weber are stated explicitly and located in the original edition; the paper does not rebuild the whole theory of elliptic functions from elementary analysis.

1 The historical question and the scope of the reconstruction

The question communicated by Ben Green through Terence Tao [M4] asks how early one could have proved that there is no tenth imaginary quadratic discriminant of class number one, using results and concepts already in the literature. It invites a form of historical research concerned with what may have been possible to prove, alongside the usual questions of who proved it and when. A reconstruction should identify both the available methods and the new insight still needed to connect them.

Pratt [M3] proposed $N \leq 1891$, drawing attention to Weber's original book [H3], rather than just the later third volume of his *Lehrbuch der Algebra* cited by Stark [M2]. The relevant class invariants and examples occur in §§92–95 of the 1891 edition. Our purpose is to make that proposal concrete and reviewable, particularly at the step where Weber's proof requires repair.

We use the following historical criterion. Substantial general results and organizing methods must have been published by the proposed date. New applications, deductions for the particular problem, and completions of an existing argument using its existing method are allowed. Merely deriving a substantial later method from older foundations does not backdate that method. This last distinction disqualifies the determinant route in an earlier reconstruction of this project as an answer to the present historical question; that route is not used here.

Heegner's passage from modular equations to a Diophantine restriction [M1] remains the decisive new observation in the reconstruction. It is not represented as an observation published in 1891. The case for historical eligibility is that Weber already developed the particular class invariants,

their degree reductions, and the factorization calculations to which this observation applies. We return to the limits of this claim in Section 9.

1.1 Mathematical statement

Write $D = -q < 0$ for a fundamental discriminant, and let $h(D)$ be the number of proper equivalence classes of primitive positive definite forms

$$Ax^2 + Bxy + Cy^2, \qquad B^2 - 4AC = D.$$

Proper equivalence means an integral substitution of determinant one. The form convention is explained directly below, so no theory of ideals is required for the arithmetic reductions.

Theorem 1. *Using the classical modular facts stated in Section 3, the argument below proves*

$$h(-q) = 1 \quad \Longleftrightarrow \quad q \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

For a hypothetical additional discriminant with $q > 163$, it first gives $q < 196$. All the imported modular facts are in Weber's 1891 book; the class-invariant step requiring repair is proved in Sections 4–5.

The proof is self-contained after the explicitly stated classical modular toolkit: it supplies the special modular-equation repair, the algebraic deduction, the Diophantine descent, and the finite arithmetic. It makes no use of a modern reciprocity theorem, class-field existence, the general irreducibility of class polynomials, or a theorem on integral points on elliptic curves. References marked H are historical inputs published by 1891. References marked M supply later provenance, interpretation, or expository guidance. These labels do not themselves establish historical eligibility; the source comparison in Section 9 does that work.

2 The elementary form obstruction

The small-prime obstruction emphasized by Silberman in the current discussion [M6] has a direct formulation in the quadratic forms of Gauss [H1]. In modern language, a prime that splits or ramifies must be the norm of an element when the class number is one; the imaginary part of that element forces the prime to be large. Here is the form-theoretic argument, including the exceptional even cases.

If $D = -4m$ is fundamental, then m is squarefree and $m \equiv 1, 2 \pmod 4$. Its principal form is $x^2 + my^2$. For $m \equiv 2 \pmod 4$, the primitive form $(2, 0, m/2)$ represents 2; for $m \equiv 1 \pmod 4$, use $(2, 2, (m+1)/2)$. If there is only one class, the principal form represents 2, which is impossible for $m > 2$. Thus an additional discriminant has q odd, squarefree, and $q \equiv 3 \pmod 4$.

Its principal form is

$$Q_q(u, v) = u^2 + uv + \frac{q+1}{4}v^2 = (u + v/2)^2 + qv^2/4. \tag{1}$$

Suppose an odd prime p satisfies $B^2 \equiv -q \pmod p$. Choose B odd. Then

$$\left(p, B, \frac{B^2 + q}{4p}\right)$$

is an integral positive definite form representing p. It is primitive: either $p \nmid B$, or squarefreeness of q implies $p \nmid (B^2+q)/(4p)$. Class number one forces $p = Q_q(u,v)$; since a prime is not a square, $v \neq 0$, and

$$p \geq q/4. \tag{2}$$

If $q \equiv 7 \pmod 8$, the primitive form $(2, 1, (q+1)/8)$ represents 2, giving the same contradiction for $q > 8$. Therefore

$$q > 163,\ h(-q) = 1 \quad \Longrightarrow \quad q \text{ prime}, \quad q \equiv 3 \pmod 8, \quad 3 \nmid q. \tag{3}$$

Indeed, a composite squarefree $q > 16$ has a prime divisor $p \leq \sqrt{q} < q/4$, contrary to (2).

3 The classical modular toolkit

For $z \in \mathfrak{H} = \{z : \operatorname{Im} z > 0\}$, set

$$f(z) = e^{-\pi i z/24} \prod_{n\geq 1} (1 + e^{(2n-1)\pi i z}),$$
$$f_1(z) = e^{-\pi i z/24} \prod_{n\geq 1} (1 - e^{(2n-1)\pi i z}),$$
$$f_2(z) = \sqrt{2}\, e^{\pi i z/12} \prod_{n\geq 1} (1 + e^{2n\pi i z}).$$

We use the classical normalization $j(z) = e^{-2\pi i z} + 744 + \cdots$, and the holomorphic cube root

$$\gamma_2(z) = \frac{f(z)^{24} - 16}{f(z)^8}, \qquad \gamma_2(z)^3 = j(z).$$

This branch is real and positive on the positive imaginary axis. The defining products converge, and f has no zero in $\mathfrak{H}$. Weber's product and transformation identities give

$$f(z)^8, \quad -f_1(z)^8, \quad -f_2(z)^8 \quad \text{as the roots of} \quad X^3 - \gamma_2(z) X - 16. \tag{4}$$

Here are the precise background facts imported from Weber. They are statements of nineteenth-century modular theory, not assumptions about the class-number-one classification.

1. **Modular functions and transformation laws.** The function j separates orbits of $\mathrm{SL}_2(\mathbb{Z})$. The function $U = f^{24}$ separates orbits of

$$\Gamma_\theta = \langle S, T^2 \rangle, \qquad S(z) = -1/z, \quad T(z) = z + 1.$$

Modulo 2, the matrices of Γ_θ are I and $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The modular functions for Γ_θ are rational functions of U. Its two cusps have $U = 0$ and $U = \infty$, represented by 1 and $i\infty$. Consequently a modular function for Γ_θ that is holomorphic on $\mathfrak{H}$ and finite at the first cusp is a polynomial in U, provided it is meromorphic at the second cusp. These facts are in §§49 and 76, pp. 147–150 and 267–268.

For later use put $F = f^3$, and $\zeta_8 = e^{\pi i/4}$. Then

$$F(Sz) = F(z), \qquad F(z+2) = \zeta_8^{-1} F(z). \tag{5}$$

For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_\theta$, Weber's formula in §35, p. 107, after cubing, is

$$F(\gamma z) = e^{-\pi i L(\gamma)/8} F(z), \qquad L(\gamma) = db + ca + 2d^2 - 2dc - 2da \pmod{16}. \tag{6}$$

In particular the multiplier is 1 if a, d are divisible by 16.

2. **The class polynomial.** If z is the upper-half-plane root of a primitive integral quadratic equation of negative discriminant D, then $j(z)$ is an algebraic integer satisfying a monic polynomial in $\mathbb{Z}[X]$ of degree $h(D)$. In particular $h(D) = 1$ implies $j(z) \in \mathbb{Z}$. We need only this degree bound, not irreducibility. The integrality proof is in §87 and the class-polynomial proof in §88, pp. 331–337. The latter uses an induction and polynomial greatest common divisors, not a class-field theorem.

3. **The level-two equation.** There is a polynomial $\Phi_2(X, Y) \in \mathbb{Z}[X, Y]$, monic of degree three in X, with
$$\Phi_2(X, j(z)) = (X - j(2z))(X - j(z/2))(X - j((z + 1)/2)). \tag{7}$$
See §72, pp. 247–255, and its use in §89, p. 338.

4. **The cube-root class invariant.** If z satisfies the primitive equation $Az^2 + Bz + C = 0$, where $3 \nmid A(B^2 - 4AC)$ and $3 \mid B$, then
$$\gamma_2(z) \in \mathbb{Q}(j(z)). \tag{8}$$
This is the theorem of §92, pp. 350–352, using the transformation equations of §74. Its proof selects one of the three cube roots of $j(z)$ by the congruence $3 \mid B$, then takes a polynomial greatest common divisor. This is a different, already established class invariant from the square invariant discussed next.

Weber's §94, Theorems I and II, asserts the additional square invariant needed below. We do *not* simply import its proof: Sections 4–5 supply the missing root control. The stronger cube-invariant assertion stated conjecturally on p. 365 is never used. Appendix C records the distinction and the normalization of (6).

4 A modular equation with exactly the required roots

Stark [M2, §3] identified the difficulty with Weber's chosen Schläfli equation: an algebraic relation alone does not ensure that all its roots come from the intended modular transforms. Additional roots would invalidate the subsequent greatest-common-divisor argument. His remedy is to construct the polynomial as a product over those transforms. The following is a specialization to an odd prime p, sufficient for our purpose. The construction is the one described in Weber's §76, pp. 267–269, with the rationality argument made explicit.

Define
$$M_\infty z = pz, \qquad M_\nu z = \frac{z + 16\nu}{p} \quad (0 \le \nu < p).$$
For $e = 1$ or 8, form
$$\prod_{M \in \{M_\infty, M_0, \dots, M_{p-1}\}} (X - F(Mz)^e F(z)^{23pe}) = \Sigma_{p,e}(X, U(z)). \tag{9}$$

Lemma 2. *The right side of (9) is a polynomial $\Sigma_{p,e}(X, U) \in \mathbb{Q}[X, U]$. Set*
$$R_{p,e}(X) = \Sigma_{p,e}(X^{23p+1}, X^{8/e}).$$
This is a nonzero polynomial in $\mathbb{Q}[X]$. For every $w \in \mathfrak{H}$,
$$R_{p,e}(F(w)^e) = 0 \quad \Longleftrightarrow \quad F(Mw)^e = F(w)^e \text{ for at least one } M. \tag{10}$$

Proof. We first verify that the factors in (9) are permuted by T^2 and S. Under $z \mapsto z+2$, choose ν' with $16\nu' \equiv 16\nu + 2 \pmod{p}$. Then

$$M_\nu(z+2) = M_{\nu'} z + \ell, \qquad \ell = (16\nu + 2 - 16\nu')/p \equiv 2p \pmod{16}.$$

Here ℓ is even; the last congruence follows from $p^2 \equiv 1 \pmod 8$. Thus $F(M_\nu(z+2)) = \zeta_8^{-p} F(M_{\nu'} z)$. The same multiplier occurs for M_∞. After taking the e-th power, it cancels the multiplier ζ_8^{-23pe} of $F(z)^{23pe}$, since $24pe$ is divisible by 8.

Under S, the transforms M_∞ and M_0 exchange. For $0 < \nu < p$, choose $0 < \nu' < p$ with $256\nu\nu' \equiv -1 \pmod{p}$. Then

$$M_\nu S = \gamma M_{\nu'}, \qquad \gamma = \begin{pmatrix} 16\nu & (-1 - 256\nu\nu')/p \\ p & -16\nu' \end{pmatrix} \in \Gamma_\theta.$$

Its diagonal entries are divisible by 16, so (6) gives $F(\gamma z) = F(z)$. This proves the claimed permutation.

The coefficients of the product are therefore modular functions for Γ_θ, holomorphic on $\mathfrak{H}$. They are finite at the cusp 1: each $M(1)$ is a rational number with odd numerator and odd denominator after reduction, so is equivalent to that cusp, where F tends to zero. Their product expansions show that they are meromorphic at infinity. The first toolkit fact now makes each coefficient a polynomial in U.

To check its numerical coefficients, put $Q = e^{\pi i z/8}$. Then

$$F(z) = Q^{-1} \prod_{n \geq 1} (1 + Q^{8(2n-1)})^3 \in \mathbb{Z}((Q)).$$

In $F(M_\nu z)$, replace Q by $\zeta_p^\nu Q^{1/p}$, where $\zeta_p = e^{2\pi i/p}$. Every automorphism $\zeta_p \mapsto \zeta_p^a$ permutes the factors indexed by ν; the remaining factor has rational coefficients. Thus the coefficient series of (9) are rational. Invariance under T^2 makes them series in integral powers of Q^8. Since $U = Q^{-8}(1 + \cdots)$ has rational coefficients and leading coefficient 1, successive subtraction of its powers shows that the polynomial coefficients belong to $\mathbb{Q}$.

Substitute $X = F(w)^{e(23p+1)}$ in (9). Since $F(w) \neq 0$, factoring out $F(w)^{23pe}$ from each factor gives exactly (10). Finally the powers of Q at infinity show that no $F(Mz)^e$ is identically $F(z)^e$: their leading exponents are multiplied by p or $1/p$. Their finite product of differences is not identically zero, so $R_{p,e}$ is not the zero polynomial. □

Only rational coefficients are needed. The construction in fact permits an integrality refinement, but it contributes nothing to the proof. The purpose of the product is the equivalence (10), which rules out precisely the extra roots at issue in Weber's proof.

5 Recovering the square class invariant

From now on let q satisfy (3), and put

$$\theta = i\sqrt{q}, \qquad J = j(\theta).$$

We prove the special case of Weber's assertion required here:

$$f(\theta)^2 \in \mathbb{Q}(J). \tag{11}$$

5.1 The twenty-fourth power

By (4), the three roots of

$$(X-16)^3 - JX = 0 \tag{12}$$

are

$$u_0 = f(\theta)^{24}, \quad u_1 = f(\theta+1)^{24}, \quad u_2 = f(1-1/\theta)^{24}.$$

We claim that only u_0 is a root of $R_{q,8}$. It is a root because $M_0\theta = \theta/q = -1/\theta$.

For the exclusion, suppose w satisfies a primitive equation $Aw^2+2Bw+C=0$ with $AC-B^2=q$, and suppose $f(Mw)^{24} = f(w)^{24}$. The orbit-separation property gives $Mw = \gamma w$ for $\gamma \in \Gamma_\theta$. The integral matrix $\gamma^{-1}M$, of determinant q, fixes w, so is of the form

$$\begin{pmatrix} h - B\ell & -C\ell \\ A\ell & h + B\ell \end{pmatrix}, \qquad h, \ell \in \mathbb{Z}, \qquad h^2 + q\ell^2 = q. \tag{13}$$

The integrality of ℓ follows by comparing the fixed-point quadratic with the primitive equation; then h is integral too. As q is not a square, necessarily $h = 0, \ell = \pm 1$. Every M is the identity modulo 2. Consequently the matrix in (13), modulo 2, must be either I or $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

For $w = \theta+1$, its coefficients are $(A,B,C) = (1,-1,q+1)$; for $w = 1-1/\theta$, they are $(q,-q,q+1)$. In either case the matrix is $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ modulo 2, which is forbidden. The claim follows from (10).

The root u_0 is simple in (12). For example, its derivative there is $(u_0-16)^2(2+16/u_0) > 0$; the product gives $u_0 > 16$ in our range. Taking the monic greatest common divisor of (12) and $R_{q,8}$ over $\mathbb{Q}(J)$ therefore gives $X - u_0$. Hence

$$f(\theta)^{24} \in \mathbb{Q}(J). \tag{14}$$

5.2 The sixth power

The eight roots of $X^8 - f(\theta)^{24}$ are

$$F(\theta + 2j) = \zeta_8^{-j} F(\theta), \qquad 0 \le j < 8.$$

We determine which are also roots of $R_{q,1}$. Write $w = \theta+2j$, $B = -2j$, so that $w^2+2Bw+(q+B^2) = 0$. As above, equality of the F values first implies equality of the f^{24} values. Formula (13) then applies with $h = 0, \ell = \pm 1$. The transform M_∞ is impossible because the lower-left entry of the resulting γ would be $\pm 1/q$. For M_ν, integrality forces $16\nu \equiv B \pmod q$. For this unique ν, up to an irrelevant overall sign,

$$\gamma = \begin{pmatrix} a & 1+Ba \\ -1 & -B \end{pmatrix}, \qquad a = (B - 16\nu)/q.$$

Since B is even and $q \equiv 3 \pmod 8$, $a \equiv 3B \pmod{16}$. Substitution in (6) gives

$$L(\gamma) = -3B + 2B^2 - a(B-1)^2 \equiv 6B \equiv -12j \pmod{16}.$$

For clarity, the last congruence follows on putting $B = 2b$: $L - 6B$ is congruent to $-8b(3-4b+3b^2)$, divisible by 16. Thus the multiplier is $(-i)^j$, and equality of the F values holds exactly for $j = 0, 4$. These give $F(\theta)$ and $-F(\theta)$.

The monic greatest common divisor of $X^8 - f(\theta)^{24}$ and $R_{q,1}$ is consequently $X^2 - f(\theta)^6$. Together with (14), this proves

$$f(\theta)^6 \in \mathbb{Q}(J). \tag{15}$$

Finally, the cube-root class-invariant theorem (8) applies to $\theta^2+q=0$. It gives $\gamma_2(\theta)\in\mathbb{Q}(J)$, and this value is nonzero by its product expression in our range. Therefore

$$f(\theta)^8=\frac{f(\theta)^{24}-16}{\gamma_2(\theta)}\in\mathbb{Q}(J),\qquad f(\theta)^2=\frac{f(\theta)^8}{f(\theta)^6}\in\mathbb{Q}(J).$$

This proves (11). It is the combination of Weber's eighth- and sixth-power results on pp. 360–362; it uses no conjectural statement about $f(\theta)$ itself.

6 Heegner's cubic comparison

The step in this section is Heegner's [M1], as explained by Stark [M2, §2]. We include every algebraic calculation and use a rational-root argument to avoid the general class-polynomial irreducibility theorem.

Put

$$\tau=\frac{3+\theta}{2},\qquad g=\gamma_2(\tau),\qquad \alpha=\frac{2}{f(\theta)^2}.$$

The equation of τ is $\tau^2-3\tau+(q+9)/4=0$, of discriminant $-q$. The class polynomial gives $j(\tau)\in\mathbb{Z}$, and (8) gives $g\in\mathbb{Q}$. As $g^3=j(\tau)$, the rational-root theorem gives $g\in\mathbb{Z}$. Since $2\tau=3+\theta$, (7) gives $[\mathbb{Q}(J):\mathbb{Q}]\le 3$. In view of (11),

$$[\mathbb{Q}(\alpha):\mathbb{Q}]\le 3. \tag{16}$$

Weber's product identity in §29, used explicitly in §95, pp. 367–368, gives

$$\alpha=e^{-\pi i/4}f_2(\tau)^2. \tag{17}$$

It can also be checked directly: for $t=e^{\pi i\theta}$, separating even and odd indices gives

$$\prod_{n\ge1}(1+(-t)^n)\prod_{n\ge1}(1+t^{2n-1})=1.$$

Substitution in the definitions of f and f_2 gives exactly (17), including its phase and factor 2. In particular $\alpha^4=-f_2(\tau)^8$; applying (4) at τ yields

$$\alpha^{12}-g\alpha^4-16=0. \tag{18}$$

Thus α is an algebraic integer. The factor 2 in its definition and the phase in (17) are both essential. The product at the purely imaginary point θ also gives

$$0<\alpha<2e^{-\pi\sqrt{q}/12}<1. \tag{19}$$

Set $t=\alpha^4$. Then $0<t<1$ and

$$P(X)=X^3-gX-16,\qquad g=t^2-16/t<0.$$

The polynomial P is strictly increasing on the real line, so its only real root is $t\in(0,1)$. A reducible monic integral cubic has an integer root; P has none. Hence it is irreducible, and

$$[\mathbb{Q}(\alpha^4):\mathbb{Q}]=[\mathbb{Q}(\alpha):\mathbb{Q}]=3. \tag{20}$$

This proves the particular degree statement needed without importing a theorem that all singular moduli have degree equal to their class number.

Choose $\beta = \alpha$ or $-\alpha$, and write its monic minimal polynomial as

$$X^3 + aX^2 + bX + c, \qquad a, b, c \in \mathbb{Z}.$$

Squaring the roots produces a cubic with coefficients

$$e = 2b - a^2, \qquad f = b^2 - 2ac, \qquad k = -c^2.$$

Squaring again produces

$$X^3 + (2f - e^2)X^2 + (f^2 - 2ek)X - k^2.$$

By (20), this is the minimal polynomial P of $\beta^4 = \alpha^4$. Comparing constant terms gives $c^4 = 16$. Change the sign of β if necessary so that $c = 2$. The coefficient of X^2 then gives

$$2(b^2 - 4a) = (2b - a^2)^2. \tag{21}$$

Modulo 2, this implies a even; modulo 4 it then implies b even. Set

$$x = -a/2, \qquad y = (b - a^2)/2.$$

Equation (21) becomes

$$y^2 = 2x(x^3 + 1). \tag{22}$$

Pratt's version $y^2 = 2x(x^3 - 1)$ is the same equation after $x \mapsto -x$.

7 Solving the Diophantine equation

Lemma 3. *The integral solutions of* (22) *are*

$$(x, y) = (0, 0), \quad (-1, 0), \quad (1, \pm 2), \quad (2, \pm 6).$$

Proof. Deal first with $x = 0, -1$. Otherwise $\gcd(x, x^3 + 1) = 1$; in the product $2x(x^3 + 1)$, each odd prime has an even exponent. The signed squarefree part of $x^3 + 1$ is therefore one of $1, -1, 2, -2$, giving four cases.

If $x^3 + 1 = z^2$, Euler's descent gives $x = -1, 0, 2$. Euler published this result in 1747 [H2, Theorem 10]; Appendix A supplies the descent, including the decrease and the terminal case.

If $x^3 + 1 = -z^2$, put $m = -x$, so $m^3 = z^2 + 1$. If z is odd, this would be $m^3 \equiv 2 \pmod 8$, impossible. Otherwise m is odd. In $\mathbb{Z}[i]$ the factors $z + i, z - i$ are coprime: a common prime would divide $2i$ and the odd norm m^3. Unique factorization therefore gives $z + i = (a + bi)^3$; units can be absorbed into cubes. Imaginary parts give $b(3a^2 - b^2) = 1$, forcing $a = 0, b = -1$. Thus $z = 0, m = 1$, the already treated solution $x = -1$.

If $x^3 + 1 = 2z^2$, equation (22) forces $x = w^2$ for an integer w. Then

$$(w^2 + 1)(w^4 - w^2 + 1) = 2z^2.$$

The factors are coprime: a common divisor divides 3, but 3 never divides $w^2 + 1$. The second factor is odd, so must be a square. For $|w| \geq 2$, however,

$$(w^2 - 1)^2 < w^4 - w^2 + 1 < (w^2)^2.$$

The value $w = 0$ fails, and $w = \pm 1$ gives $x = 1$.

If $x^3 + 1 = -2z^2$, put $m = -x$. Now $m^3 = 1 + 2z^2$ is odd. The factors $1 \pm z\sqrt{-2}$ are coprime in $\mathbb{Z}[\sqrt{-2}]$: a common prime would divide 2 and an odd norm. Unique factorization gives $1 + z\sqrt{-2} = (a + b\sqrt{-2})^3$, since either unit is a cube. Real parts yield $a(a^2 - 6b^2) = 1$, forcing $a = 1, b = 0$. Again $x = -1$.

For completeness, the two unique-factorization assertions here need only the Euclidean algorithm in these specific rings. Given a quotient in $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-2})$, round its two coefficients to integers. The error has norm at most $1/2$ or $3/4$, respectively. Thus division has a remainder of strictly smaller norm. The Euclidean algorithm supplies greatest common divisors and Bezout identities; hence irreducibles are prime. Factoring by decreasing positive integer norms proves existence and uniqueness. The units are $\{\pm 1, \pm i\}$ and $\{\pm 1\}$. These are explicit Euclidean factorizations, with no invocation of ideal class theory. Substituting the resulting values of x gives the asserted list. □

Reversing the substitution in Section 6, the possible pairs of cubic coefficients are

$$(a, b) = (0, 0), (2, 4), (-2, 0), (-2, 8), (-4, 4), (-4, 28). \tag{23}$$

In particular, $|a| \le 4$ and $0 \le b \le 28$.

8 The bound and the finite finish

The cubic equation of $\beta = \pm\alpha$, with constant term 2, and (23) imply

$$2 = |\beta^3 + a\beta^2 + b\beta| \le \alpha^3 + 4\alpha^2 + 28\alpha < 33\alpha.$$

Together with (19) this gives

$$e^{\pi\sqrt{q}/12} < 33, \qquad \sqrt{q} < \frac{12 \log 33}{\pi} < 14, \qquad q < 196.$$

For the last inequality it is enough to use $\pi > 3$ and $e^{7/2} > 33$. If an exact arithmetic check of the latter is desired, the exponential series through degree 11 is

$$\sum_{n=0}^{11} \frac{(7/2)^n}{n!} = \frac{386627544403}{11678515200} > 33.$$

Only $171, 179, 187, 195$ lie strictly between 163 and 196 and are 3 modulo 8. Each has a witness to the obstruction (2):

q	171	179	187	195
p	3	3	11	3
r	0	1	0	0

$p \mid q + r^2, \quad 4p < q.$

This contradicts class number one in every case. One may instead discard the three composite values immediately and use only the witness $p = 3$ for 179. Appendix B gives the small remaining classification below 164 and checks the nine survivors by reduction. No singular modulus has to be evaluated numerically.

9 What the date 1891 does and does not mean

The evidence for this cutoff is more specific than a list of old foundations. Weber's §76, pp. 267–269, explicitly constructs transformation equations by taking functions permuted under $z \mapsto z+2$ and $z \mapsto -1/z$, and expressing their symmetric functions through f^{24}. His §94 gives the eighth- and sixth-power invariants and explicitly combines them to obtain the square invariant. Section 4 repairs the root control by staying within that same product construction. The cyclotomic coefficient argument uses the familiar action of automorphisms on roots of unity; it is not a general reciprocity law.

There is also direct evidence for the final algebraic setting. On p. 368 Weber factors the equations for the known examples through degrees 24, 12, 6, and 3, and compares their coefficients. What he does not do is turn the coefficient restrictions into a Diophantine equation that exhausts all possibilities. That is Heegner's observation. Stark's retrospective assessment [M2, §5] is that Weber could have completed the argument had he noticed this consequence. Pratt's bibliographical contribution [M3] is to locate the material in 1891, rather than stopping at the later edition Stark cited.

Under the criterion in Section 1, we therefore regard $N \leq 1891$ as a defensible proposal: the substantial modular methods are already present, the repair is an application of one of them, and the remaining new step concerns the particular class-number-one problem. This is a judgment about admissibility, supported by the displayed construction. If the criterion were to require every application and every repair itself to have appeared in print by the cutoff, this reconstruction would not satisfy it. That stricter rule would answer a different question from whether a proof could have been assembled using the available methods.

Nor does the argument show that discovery in 1891 was likely. A possible reason for the delay is that calculating striking singular values and proving a uniform classification suggest different directions of work: Weber starts with known discriminants and derives their small equations; Heegner uses the necessary form of those equations to rule out unknown discriminants. This reversal is visible in the mathematics, but an explanation of why a particular mathematician did not make it would require evidence from that mathematician's work and correspondence. We offer the contrast as a question for historical research, not as an estimated probability of discovery.

Finally, no earliest year is proved. The present source audit identifies 1891 as a documented point at which the needed material is together. It does not rule out an earlier route or an earlier publication of enough of Weber's ingredients. The credit for the actual theorem and its later clarification remains with Heegner and the mathematicians who completed and established its proofs.

A Euler's descent, with the decrease made explicit

Euler's Theorem 10 in [H2], written in 1738 and published in 1747, treats a cube plus one that is a square. We reproduce the descent in a form that also covers zero and negative rational inputs. Conrad's exposition [M5] is useful guidance for making the sign choices, the descent measure, and the terminal case explicit. These details elaborate Euler's factorization and rational parametrizations; they use no theory of elliptic curves.

Lemma 4. *The rational solutions of* $Y^2 = X^3 + 1$ *are*

$$(-1, 0), \quad (0, \pm 1), \quad (2, \pm 3).$$

Proof. First establish the descent assertion: if u, v are coprime positive integers, $3 \nmid u$, and

$$u,\ v,\ u^2 - 3uv + 3v^2 \quad \text{are squares,} \tag{24}$$

then $u = v = 1$.

Write $w^2 = u^2 - 3uv + 3v^2$, choosing the sign so that $w \equiv -u \pmod 3$. Put

$$\frac{w-u}{v} = \frac{m}{n}, \qquad \gcd(m,n) = 1, \quad n > 0.$$

Here $3 \nmid m$ and $n \mid v$. Squaring $w = u + (m/n)v$ and simplifying gives

$$\frac{u}{v} = \frac{3n^2 - m^2}{n(2m+3n)}.$$

The fraction on the right is reduced. A prime common to $3n^2 - m^2$ and $2m + 3n$ would divide both m and $3n$, contradicting $\gcd(m,n) = 1$ and $3 \nmid m$; the other denominator factor n is also coprime to the numerator. Since u is a square not divisible by 3, comparing signs modulo 3 forces

$$u = m^2 - 3n^2, \qquad v = -n(2m+3n). \tag{25}$$

Choose k with $k^2 = u$ and $k \equiv -m \pmod 3$, and reduce the nonzero fraction

$$\frac{k-m}{n} = \frac{u'}{v'}, \qquad \gcd(u',v') = 1, \quad v' > 0.$$

It has $3 \nmid u'$. From $m^2 - 3n^2 = (m + (u'/v')n)^2$, one obtains

$$v = n^2 \frac{u'^2 - 3u'v' + 3v'^2}{u'v'}. \tag{26}$$

The quadratic in the numerator is positive. Since $v > 0$ and $v' > 0$, also $u' > 0$. The three positive integers $u', v', u'^2 - 3u'v' + 3v'^2$ are pairwise coprime: the only possible additional common prime would be 3 dividing u'. As v is a square, multiplying (26) by $(u'v'/n)^2$ shows that their product is a rational square, hence an integer square. Each is therefore a square. This constructs another pair satisfying (24).

Moreover, v/n is an integer by (25). Equation (26), together with the coprimality just proved, implies $u'v' \mid n$. Thus

$$0 < v' \le u'v' \le n \le v.$$

If $v' = v$, every inequality is equality, so $u' = 1$, $v' = n = v$. Then $(k-m)/n = 1/n$, or $k = m + 1$. Combining $m^2 - 3n^2 = k^2$ with $v = n = -n(2m+3n)$ gives $2m + 1 = -3n^2 = -3n$, hence $n = 1, m = -2, u = v = 1$. If instead the new pair is already $(u', v') = (1, 1)$, then $k = m + n$, so $m = -2n$. Coprimality of m, n again forces $n = 1$ and $u = v = 1$. Consequently any nontrivial pair descends to a nontrivial pair with strictly smaller positive v, which is impossible. This proves the descent assertion.

Now let $Y^2 = X^3 + 1$. Necessarily $X \ge -1$, and $X = -1$ gives $Y = 0$. Otherwise write $X = a/b$, $b > 0$, $\gcd(a,b) = 1$, and set $c = a + b > 0$. The equation implies that

$$b(a^3 + b^3) = bc(c^2 - 3bc + 3b^2)$$

is an integer square. If $3 \nmid c$, the three factors are pairwise coprime and positive, so are squares. Apply the descent assertion with $(u, v) = (c, b)$ to get $b = c = 1$, and thus $X = 0$.

If $c = 3d$, then $3 \nmid b$, and after removing a factor 9 the square condition is

$$bd(b^2 - 3bd + 3d^2) \text{ is a square.}$$

The factors are again pairwise coprime. The descent assertion with $(u, v) = (b, d)$ gives $b = d = 1$, hence $c = 3, a = 2, X = 2$. The corresponding values of Y complete the list. □

B The small discriminants and the nine survivors

We supply the classification below 164 rather than referring to an earlier paper. Section 2 excludes every even fundamental discriminant except $-4, -8$, and for odd $q > 8$ forces $q \equiv 3 \pmod 8$. The value 15 is excluded by the ramified prime 3; every composite $q > 16$ is excluded as in (3). The primes between 9 and 163 congruent to 3 modulo 8 are

$$11, 19, 43, 59, 67, 83, 107, 131, 139, 163.$$

The four values $59, 83, 107, 131$ have $-q \equiv 1 \pmod 3$, and 139 has $-139 \equiv 1 \pmod 5$. The primes 3 and 5 are below $q/4$, excluding these values by (2). Including the smaller fundamental discriminants leaves exactly the nine in Theorem 1.

To check that they do have class number one, use elementary reduction of positive definite forms [H1]. Replacing B by its residue modulo $2A$ makes $|B| \le A$; if $C < A$, the substitution $(x, y) \mapsto (-y, x)$ exchanges the outer coefficients and strictly decreases the positive first coefficient. Iteration therefore gives

$$|B| \le A \le C, \qquad 3A^2 \le 4AC - B^2 = q.$$

On the boundary, take $B \ge 0$ when $|B| = A$ or $A = C$. The usual substitutions identify the two signs there. For each of the nine values, the finite check of $B^2 \equiv -q \pmod{4A}$ gives the following complete table:

q	$\lfloor\sqrt{q/3}\rfloor$	Reduced forms (A, B, C)
3	1	$(1, 1, 1)$
4	1	$(1, 0, 1)$
7	1	$(1, 1, 2)$
8	1	$(1, 0, 2)$
11	1	$(1, 1, 3)$
19	2	$(1, 1, 5)$
43	3	$(1, 1, 11)$
67	4	$(1, 1, 17)$
163	7	$(1, 1, 41)$

Every form is equivalent to a reduced form, and each row has just one. This verifies the other implication of Theorem 1. The finite work consists only of these small divisions and congruences.

C Source audit and the remaining historical judgment

All page references below are printed pages of the 1891 edition, not PDF page numbers or page numbers in the 1908 edition.

Input or method	Weber (1891)	Role in this paper
Product definitions and identities	§29; §49, pp. 149–150	Defines the functions and the cubic (4).
Linear transformation of f	§35, pp. 105–107	After cubing, gives the exact multiplier (6).
Functions for the theta subgroup	§49, pp. 147–149; §76, pp. 267–268	Converts invariant coefficient functions into polynomials in f^{24}.
Invariant modular equations	§72, pp. 247–255	Supplies the degree-three level-two equation.
Products over modular transforms	§§73,76, pp. 255–269	The existing organizing method used in the root-control repair.
Integrality and the class polynomial	§§87–88, pp. 331–337	Only degree at most the class number is needed.
γ_2 as a class invariant	§92, pp. 350–352	Applied separately at θ and τ.
f^8, f^6, and their quotient f^2	§94, pp. 360–362	Statements located; the special case needed is proved in Section 5 with the repaired equation.
Stronger f^3 assertion	§94, p. 365	Explicitly conjectural there; not used.
Low-degree factorizations for known examples	§95, pp. 367–368	Evidence that the specific algebraic setting was already under study.

The multiplier convention. Weber writes the transform as $(\gamma + \delta z)/(\alpha + \beta z)$. His final formula on p. 107 has a cube-root-of-unity factor, a factor $(2/(\alpha - \beta))$, and the exponential $\exp[-3\pi i(\alpha-\beta)(\alpha+\beta+\gamma-\delta)/8]$. Upon cubing, the first factor disappears. Using $(2/r) = (-1)^{(r^2-1)/8}$ for odd r and $\alpha\delta - \beta\gamma = 1$ gives (6) after $(a, b, c, d) = (\delta, \gamma, \beta, \alpha)$. This also checks S and T^2 directly and avoids confusing Weber's matrix convention with the modern one.

What was actually checked. The 1891 statements and the relevant transformation and product-method passages were compared with the original scan. Stark's full 1969 paper, especially pp. 20–25, was used to identify the root-control gap and its repair. The specialization above gives explicit transformations and congruences for an odd prime, rather than treating Stark's later paper as a historical black box. Euler's Theorem 10 was located in the original work and its German translation; its publication date 1747 is recorded in the Euler Archive. The descent above supplies the details needed for a complete proof rather than relying on Euler's brief assertion that the new numbers are smaller.

Verification status. The accompanying script checks polynomial identities, multiplier congruences, descent identities, the rational exponential inequality, and the finite form enumeration using exact arithmetic. These checks do not verify the analytic toolkit or the historical interpretation. This working draft has not been assigned an independent audit verdict and is not a formal verification. The remaining historical judgment is whether the explicitly displayed repair and Heegner's problem-specific deduction meet the criterion in Section 1; the date 1891 is advanced with that criterion visible.

References

[H1] C. F. Gauss, *Disquisitiones arithmeticae*, Leipzig, 1801, Section V, especially Articles 153–159 and 171–175. Original edition.

[H2] L. Euler, *Theorematum quorundam arithmeticorum demonstrationes*, Commentarii academiae scientiarum Petropolitanae **10** (published 1747; written 1738), 125–146, Theorem 10, Enestrom E98. Euler Archive record and translations; original scan.

[H3] H. Weber, *Elliptische Functionen und algebraische Zahlen*, Friedrich Vieweg und Sohn, Braunschweig, 1891. Original edition. This is the edition used for all section and page references here.

[M1] K. Heegner, *Diophantische Analysis und Modulfunktionen*, Mathematische Zeitschrift **56** (1952), 227–253. doi:10.1007/BF01174749.

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